G-Symmetric Monoidal

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Externalizing Tambara Functors

Internal vs External Mackey Functors

Had an equivalence between Mackey functors and extensions

$$\begin{tikzcd} \underline{\mathcal{F}in}^{\cong} \arrow[r, "T \mapsto \underline{M}_T"] \arrow[d, hook] & \underline{\mathcal{C}oeff}. \\ \underline{\mathcal{F}in} \arrow[ur, dotted] & \end{tikzcd}$$

Question

1. *How can we simplify this?*
2. *How do we fold in the twisted tensor product?*

G-Commutative Monoid

Restriction

We have a natural isomorphism

$$i_H^*(\underline{M}_T) \cong i_H^* \underline{M}_{i_H^* T}$$

Transfers Restrict

If T is an admissible G-set for $\mathcal{D}(V)$, then the restriction of

$$\underline{\pi}_k(X)_T \to \underline{\pi}_k(X)$$

is

$$i_H^*(\underline{\pi}_k(X))_{i_H^* T} \to i_H^*(\underline{\pi}_k(X)).$$

Aside: Overdetermination

Remark

If T is an H-set, then the map $\left(i_H^*\underline{M}\right)_T \to i_H^*\underline{M}$ is a retract of the restriction to H of

$$\underline{M}_{G\underset{H}{\times}T} \to \underline{M}_{G/H}.$$

Corollary

A Mackey functor is equivalently an extension

$$\begin{tikzcd}
\mathcal{F}in^{G,\cong} \arrow[r, "T\mapsto \underline{M}_T"] \arrow[d, hook] & \mathcal{C}oeff^G. \\
\mathcal{F}in^G \arrow[ur, dotted] &
\end{tikzcd}$$

G-Symmetric Monoidal Structure

Main Takeaway

Have a coefficient system of symmetric monoidal categories $\underline{\mathcal{C}}$ together with

$$N^{(-)}(-)\colon \underline{\mathcal{F}in}^{\cong} \times \underline{\mathcal{C}} \to \underline{\mathcal{C}}$$

that is "bilinear".

Prototypical Example

If $\underline{\mathcal{C}}$ is a symmetric monoidal Mackey functor, with symmetric monoidal transfer maps

$$N_K^H\colon \underline{\mathcal{C}}(G/K) \to \underline{\mathcal{C}}(G/H),$$

then we get such a structure via

$$N^{H/K}(-) = N_K^H i_K^*\colon \underline{\mathcal{C}}(G/H) \to \underline{\mathcal{C}}(G/H).$$

G-Cartesian Structure

Example

$\underline{\mathcal{C}}$ a coefficient system of Cartesian monoidal categories and N_K^H the right adjoint to the restriction i_K^*.

Example

Take $\underline{\mathcal{C}oeff}$ with the usual restriction maps and with coinduction.

Example

Take $\underline{\mathcal{T}op}$ to be G-spaces with Cartesian product and ... coinduction.

G-coCartesian Structure

Example

$\underline{\mathcal{C}}$ a cofficient system with the co-Cartesian monoidal structure and N_K^H the left adjoint to the restrictions.

Example

$\underline{\mathcal{C}oeff}$ or $\underline{\mathcal{T}op}$ with induction.

Example

Representations of H with induction = coinduction and restriction.

G-symmetric monoidal functors

Definition

A functor F on coefficient systems is G-symmetric monoidal if

1. F is symmetric monoidal and
2. $F(N^T X) \to N^T F(X)$.

Example

The homotopy coefficient systems:

$$\underline{\pi}_k : \underline{\mathcal{T}op}_* \to \underline{\mathcal{C}oeff}.$$

G-Commutative Monoids

An object $X \in \underline{\mathcal{C}}(G/G)$ gives a functor

$$N^{(-)}\underline{X} \colon \underline{\mathcal{F}in}^{\cong} \to \underline{\mathcal{C}}$$

via $T \in \mathcal{F}in^{H}$ maps to

$$N^{T} i_H^{*} X.$$

Definition

A G-commutative monoid is an object M together with an extension

$$\begin{tikzcd} \underline{\mathcal{F}in}^{\cong} \arrow[r, "T \mapsto N^T \underline{M}"] \arrow[d, hook] & \underline{\mathcal{C}oeff}. \\ \underline{\mathcal{F}in} \arrow[ur, dotted] & \end{tikzcd}$$

$\mathcal{O}$-Commutative Monoids

Proposition

If $\mathcal{O}$ is a transfer system, then the restriction on $\mathcal{F}in^G_{\mathcal{O}}$ lands in $\mathcal{F}in^H_{i_H^*\mathcal{O}}$.

Definition

An $\mathcal{O}$-commutative monoid is an object M together with an extension

$$\begin{tikzcd} \underline{\mathcal{F}in}^{\cong} \arrow[r, "T \mapsto N^T \underline{M}"] \arrow[d, hook] & \underline{\mathcal{C}oeff}. \\ \underline{\mathcal{F}in}_{\mathcal{O}} \arrow[ur, dotted] & \end{tikzcd}$$

Examples in Coefficient Systems

Theorem

An $\mathcal{O}$-commutative monoid in $\mathcal{C}oeff$ is a Mackey functor for G.

Example

An $\mathcal{O}^{tr}$-commutative monoid is a coefficient system.

Ambidexterity

Proposition

If $K \to H$, then N_H^G is also the left adjoint to the restriction i_H^* on $\mathcal{O}$-commutative monoids.

Slogan IV

The G-symmetric monoidal structure becomes the G-co-Cartesian monoidal structure on the G-commutative monoids.

The Norm (in Mackey Functors)

Definition

$$\begin{tikzcd} (\mathcal{A}^{\mathcal{O}})^{\times 2} \arrow[rr, "\underline{M}\otimes\underline{N}"] \arrow[d, "\times"'] & & \mathcal{A}b. \\ \mathcal{A}^{\mathcal{O}} \arrow[urr, "\underline{M}\Box\underline{N}"'] & & \end{tikzcd}$$

Definition (Mazur, Hoyer, Chan)

$$\begin{tikzcd} \mathcal{A}^{\mathcal{O}}_H \arrow[rr, "\underline{M}"] \arrow[d, "\mathrm{Map}^H(G,-)"'] & & \mathcal{S}et. \\ \mathcal{A}^{\mathcal{O}}_G \arrow[urr, "N_H^G\underline{M}"'] & & \end{tikzcd}$$

How do we compute box products?

Box Product

1. Form the tensor of the coefficient systems
2. Take the free Mackey functor
3. enforce the relations

$$a \otimes T_f(b) = T_f\big(R_f(a) \otimes b\big) \text{ and } T_f(a) \otimes b = T_f\big(a \otimes R_f(b)\big).$$

Key Example

$T, T' \in \mathcal{F}in^G$, then

$$\underline{A}_T \Box \underline{A}_{T'} \cong \underline{A}_{T \times T'}.$$

Left adjoint

Box commutes with colimits in both variables.

How do we compute norms?

Key Example

$T \in \mathcal{F}in^H$, then $N_H^G \underline{A}_T \cong \underline{A}_{\mathrm{Map}^H(G,T)}$.

Sifted Colimits

N_H^G commutes with sifted colimits like reflexive coequalizers.

Example

$\mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2$ gives a reflexive coequalizer

$$\mathbb{Z}_{\{a,b\}} \underset{[0,1]}{\overset{[2,1]}{\rightrightarrows}} \mathbb{Z} \longrightarrow \mathbb{Z}/2, \text{ giving}$$

$$\underline{A}_{\mathrm{Map}(C_2,\{a,b\})} \rightrightarrows \underline{A} \longrightarrow N_e^{C_2}(\mathbb{Z}/2).$$

Multiplicative G-commutative monoids

Theorem (Mazur, Hoyer)

G-commutative monoids in Mackey functors are Tambara functors.

Theorem (Chan)

$\mathcal{O}_m$-commutative monoids in $\mathcal{O}_a$-Mackey functors are $(\mathcal{O}_a, \mathcal{O}_m)$-Tambara functors.

Restatement

A Tambara functor is a Mackey functor $\underline{R}$ with an extension of $T \mapsto N^T\underline{R}$ to a functor

$$\mathcal{F}in^G \to \mathcal{M}ackey.$$

Consequences

Corollary

The Mackey functor norm N_H^G is the left-adjoint to the restriction i_H^* on $\mathcal{O}$-Tambara functors whenever $H \to G$ in $\mathcal{O}$.

Example

Free Green functor on the underlying:

$$\begin{tikzcd}
\mathbb{Z}[t]/(t^2-2) \arrow[d, bend right] & \oplus & \mathbb{Z}\{t_{1,0}\} \arrow[d, bend right] & \oplus & \mathbb{Z}\{t_{2,0}, t_{1,1}\} \arrow[d, bend right] & \oplus & \mathbb{Z}\{t_{3,0}, t_{2,1}\} \arrow[d, bend right] & \cdots \\
\mathbb{Z} \arrow[u, bend right] & & \mathbb{Z}\{x, \bar{x}\} \arrow[u, bend right] & & \mathbb{Z}\{x^2, x\bar{x}, \bar{x}^2\} \arrow[u, bend right] & & \mathbb{Z}\{x^3, x^2\bar{x}, x\bar{x}^2, \bar{x}^3\} \arrow[u, bend right] & \cdots
\end{tikzcd}$$

Tensoring

Rewriting Tensoring

$\mathcal{O}$-Tambara functors always has a G-coCartesian monoidal structure:

$$N_H^G := \bigotimes_{\mathcal{O}} \mathrm{Ind}_H^G.$$

Restating

If $H \to G$ in $\mathcal{O}$, then the left adjoint to the forgetful functor can be computed in the underlying additive category.

Application: Rational Tambara

Definition (Geometric Fixed Points)

$\Phi^G \underline{M} := \underline{M}(G/G)/Im(tr)$.

Proposition

$\Phi^G N_H^G \underline{M} \cong \Phi^H \underline{M}$ and $\Phi^G(\underline{M} \square \underline{N}) \cong \Phi^G(\underline{M}) \otimes \Phi^G(\underline{N})$.

Theorem (Greenlees–May, Strickland)

$$\mathcal{M}ackey_{\mathbb{Q}}^G \xrightarrow{\simeq} \mathrm{Fun}^{\otimes}(\mathcal{F}in^{G,\cong}, \mathcal{V}ect_{\mathbb{Q}})$$

$$\underline{M} \longmapsto (T \mapsto \Phi^G N^T \underline{M})$$

$$\mathcal{T}amb_{\mathbb{Q}}^G \xrightarrow[\simeq]{} \mathrm{Fun}^{\otimes}(\mathcal{F}in^{G}, \mathcal{V}ect_{\mathbb{Q}})$$

The norm

Norm

Have functor $N_H^G\colon \mathcal{S}p^H \to \mathcal{S}p^G$ satisfying

1. strong symmetric monoidal
2. commutes with sifted colimits
3. $N_H^G(\Sigma_+^\infty X) \simeq \Sigma_+^\infty \mathrm{Map}^H(G, X)$.

Proposition

If $E \in \mathcal{S}p^H_{\geqslant 0}$: $\underline{\pi}_0 N_H^G(E) \cong N_H^G \underline{\pi}_0 E$.

Proposition

$N_H^G(S^V) \cong S^{\mathrm{Ind}_H^G V}$.

$\mathcal{O}$-algebras in $\mathcal{S}p^G$

Theorem

If G/H is an admissible G-set for $\mathcal{O}$, then $\mathcal{O}$-structures an essentially unique map

$$N^{G/H}R = N_H^G i_H^* R \to R.$$

Example

If X is an $\mathcal{O}$-algebra in spaces, then $\Sigma^\infty_+ X$ is an $\mathcal{O}$-algebra and

$$N^{G/H}\Sigma^\infty_+ X \cong \Sigma^\infty_+ \operatorname{Map}(G/H, X) \to \Sigma^\infty_+ X.$$

Lifting to higher categories

G-Symmetric Monoidal ∞-categories

Heuristics:

1. Symmetric monoidal = coCartesian fibration to $\underline{\mathcal{F}in}_*$ + Segal condition
2. Symmetric monoidal Mackey functors

G-commutative monoids

$\underline{\mathcal{F}in}_{\mathcal{O}}$ gives a G-∞-operad, and $\mathcal{O}$-commutative monoids are what we think.

Example

$\mathcal{O}$-algebras in G-spectra give the $\mathcal{O}$-commutative monoids!

Wimmer's Theorem

Theorem (Greenlees–May)

The functor

$$\begin{aligned} \mathcal{S}p^G_{\mathbb{Q}} &\xrightarrow{\simeq} \mathrm{Fun}^{\otimes}(\mathcal{F}in^{G,\cong}, \mathcal{C}h_{\mathbb{Q}}) \\ E &\longmapsto (T \mapsto \Phi^G N^T E) \end{aligned}$$

Theorem (Wimmer)

The functor

$$\begin{aligned} \mathcal{C}omm^G_{\mathbb{Q}} &\xrightarrow{\simeq} \mathrm{Fun}(\mathcal{O}rb^G, \mathcal{C}omm_{\mathbb{Q}}) \\ R &\longmapsto (T \mapsto \Phi^G N^T R) \end{aligned}$$

Compact Lie

Tensoring

We have a tensoring $\mathcal{T}op^G \times \mathcal{C}omm^G \to \mathcal{C}omm^G$ with

$$G/H \otimes R \cong \mathrm{Ind}_H^G \, i_H^* R.$$

Warning

If $\dim G/H > 0$*, then* $G/H \otimes R$ *is basically never a functor of spectra.*

Example

$$S^1/C_n \otimes R = THH_{C_n}(R).$$

Motivic

Tensoring [Hu]

We have a tensoring $\mathcal{T}op^{mot}_{/k} \times \mathcal{C}omm_{/k} \to \mathcal{C}omm_{/k}$ with

$$X \otimes \mathrm{Sym}(E) \cong \mathrm{Sym}(X_+ \otimes E).$$

Warning (Bachmann–Hoyois, Elmanto)

If $X \to k$ is not étale, then $X \otimes R$ is confusing. If $X \to k$ is étale, then this is the Weyl restriction.